# An Evolutionary Algorithm Approach to Gravitational Navigation: Analyzing the Impact of Environmental Geometry on Evolution Strategy, Parameter Optimization, and Genomic Success

1st Annie Fink
*Department of EECS*
*University of Tennessee*
Knoxville, USA
afink12@vols.utk.edu

***Abstract*—Navigating complex gravitational environments is a widespread field of research that involves balancing movement of a rocket with external gravitational forces. This study takes an evolutionary algorithm approach to find successful routes between a starting point and a target via population evolution. This process examines three different environments using a simulation, which include planets with varying masses, radii, and locations in the layout. Different values of mutation and crossover are tested to determine the optimal balance between exploration and the recombination of existing information to discover successful routes. Also, this report analyzes rocket trajectories and genome configurations of successful individuals, as well as population behaviors that emerged throughout each simulation. The overall goal is to evolve the population to be more successful in path-finding as a whole, but also to find different viable paths, rather than a singular, best trajectory. This report also analyzes complex behaviors and trends that emerge from running the same evolutionary algorithm on different map configurations. This study finds that the evolutionary algorithm can produce an optimal strategy to navigate different environments. This includes the optimal rates of exploration (mutation) and genomic trait combinations (crossover), as well as values of mass and thrust that best allow populations to find success in each distinct environment. These different factors influence the rocket trajectories to balance forces from gravity and thrust to optimally navigate the gravitational fields and increase the rate of success throughout generations.**

***Index Terms*—Evolutionary Algorithms, Planetary Gravitational Fields, Agent-Based Simulation, Path-Finding, Gravitational Physics**

## I. Introduction and Motivation

Navigation through complex gravitational fields to reach a specific destination remains a significant challenge within interstellar dynamics. In the vacuum of space, spacecraft's trajectories are heavily influenced by the gravitational forces of bodies within the universe. Therefore, there are many delicate factors that can influence whether or not a successful path is established while navigating gravitational bodies. Determining a trajectory, especially with many forces acting on the object at once, is a very computationally heavy problem requiring a substantial number of calculations and time. Evolutionary algorithms are inspired by natural evolution within species, in order to find optimal solutions to complex problems [1]. The process chooses parents through tournament selection, and utilizes mutation and crossover to discover genome combinations that optimize a fitness evaluation, and evolve a population to be increasingly more successful throughout generations. In this lab, the genome consists of a mass value, a thrust value, and an initial velocity (acts as a starting direction). Different combinations of these values are explored to find specific genomes that allow a rocket to successfully find a path to a specified target. Considering the complexity of predicting these interstellar trajectories, the goal of this report is to utilize evolutionary algorithms to find specific combinations of mass, initial velocity, and thrust that allow a rocket to successfully navigate a gravitational terrain, while also analyzing how rates of mutation and crossover affect the rates of success within populations. This report also aims to discover how the optimal parameters and genomes will potentially differ among varying environments, and to analyze how these results correlate to the strategy used to produce the best performance.

Following this introduction, this paper examines similar works to this field, explains the methods and experimental setup for the simulation, analyzes results from data gathered, and finally discusses conclusions and future work.

## II. Related Work

Using an evolutionary algorithm within gravitational navigation and rockets is a widespread field with many different approaches and goals. While the purpose of this experiment was simply to find viable trajectories to a specified target, many other reports aim to also optimize the path based on travel time, fuel consumption, etc. [3] [4] [6] [8] [9] [10] [11] [12]. Additional reports also have different objectives for optimization, including reaching a specific position or landing point [7] [14], settling into a stable orbit [13], and finding the best features for different types of rocket engines [14].

Other methods involve an inner and outer layer of computation, such as using a primary GA to optimize a genome of features and a secondary GA as its fitness function to search for optimal flight paths using that particular genome [5]. Also, other approaches involve a neural network to "pilot" a rocket based on certain controls within the genome [6], using a GA to find ideal genomes based on different optimization goals and applying game theory to those individuals [12], or employing particle swarm optimization on a population given a route itinerary produced from a GA [7].

The genome within this experiment is quite simple, only involving three traits related to the rocket's mass, thrust, and initial direction. However, previous works define their genomes by different means. For instance, some reports used specific launch dates and flight times within the genome, which can determine deployment windows that lead to successful trajectories within a time constraint [3] [6] [8] [11] [12]. Additionally, the genome could contain steering and control aspects [4] [5] [13], the order of planets/landmarks to visit [9] [10], or greatly expand the number of rocket traits to include more features [14].

Furthermore, the acceleration due to gravity within this report, as well as others, is calculated using a simplified N-Body equation, where forces from multiple bodies are acting on a rocket at once, and the rocket's mass is negligible compared to the mass of the other bodies [6] [10] [12]. However, additional works only involve the need for the Two-Body equation, as they only perform calculations based on one gravitational object in the environment, such as a launching from a planet or navigating a singular large star [3] [4] [5] [8] [10] [11] [12].

These are just a few instances of how flexible and versatile the use of genetic algorithms can be within various other applications of gravitational navigation.

## III. Method/Approach/Experimental Setup

In this simulation, the environment consists of stationary obstacles, or planets, that have mass and radius values (blue circles). A population of agents, or rockets, is initialized at a starting point (red circle), and travels through the environment to reach a target (yellow ring). As the agents move, they are subject to the gravitational forces determined by the mass of each planet. Coming into contact with the obstacle or moving too far out of bounds will "kill" the agent. If a rocket gets within 10 units of the target, the agent will freeze and be deemed a "success". I tested this simulation on three different environments, pictured below. Environment I. has many planets with a mix of radii and masses in an unstructured layout. Environment II. has large, lower mass planets in a horizontally symmetric format. Environment III. only has a few planets spread throughout, but they have very high masses with small radii.

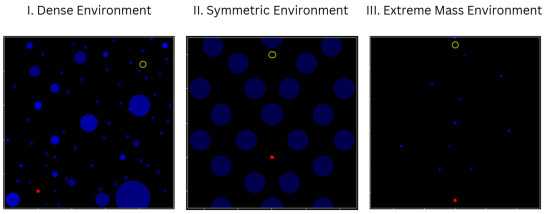



Fig. 1. Visualizations of the three unique environments tested.

Each agent is an individual within the population, capable of following their own trajectory throughout the environment. The genome of each rocket consists of three components: mass value, thrust value, and initial velocity vector. Mass and thrust remain static values that influence the rocket's acceleration throughout the entirety of its trajectory. The initial velocity acts as a starting direction for the agent. When the population is initialized, random values are assigned to each individual's genome, with the ranges being 1 to 30 for mass, 0 to 30 for thrust, and -1.5 to 1.5 for both the x and y directions of the initial velocity.

Within each generation, the agents were allowed a maximum of 5000 frames, or steps, to reach the target. In each step, each rocket $R$'s acceleration resulting from a planet $P$ is calculated using a simplified N-body vectorized gravitational acceleration formula, where the mass of the rocket is negligible compared to the mass of each planet [2]. This formula is given by:

$$\hat{a}_{grav} = G\frac{m_P}{\|r\|^2} \cdot \frac{r}{\|r\|} = G\frac{m_P(r_P - r_R)}{\|r_P - r_R\|^3} \qquad (1)$$
$$r = (x, y)$$

Where $G$ is the gravitational constant, $m_p$ is the mass of the planet, and $r_p$ and $r_r$ are the positions of the planet and rocket, respectively. Also, throughout the simulation, the planets are stationary and are never affected by the rockets. The acceleration due to thrust for each rocket is calculated using a thrust to mass ratio, along with the unit vector of its current velocity ($v$).

$$\hat{a}_{thrust} = \frac{F_{thrust}}{m_R} \cdot \frac{\hat{v}}{||\hat{v}||} \qquad (2)$$

The resultant vector produced from taking the summation of the gravitational acceleration from each planet will indicate the overall direction and magnitude that the rocket is being pulled toward. Therefore, the total acceleration vector, including its magnitude and direction, is its thrust acceleration added with this summation of all $\hat{a}_{grav}$ between the rocket and all $n$ planets.

$$\hat{a}_{total} = \hat{a}_{thrust} + \sum_{i=1}^{n} \hat{a}_{grav,i} \qquad (3)$$

Using the calculated acceleration vector, the rocket's velocity $v$ and positions $p$ are updated for the next time step:

$$\hat{v}_{t+1} = \hat{v}_t + \hat{a}_{total}\ dt \tag{4}$$

$$p_{t+1} = p_t + \hat{v}_{t+1}\ dt \tag{5}$$

The term $dt$ signifies the discrete time step. This computation is done each frame for all rockets and updates their position within the simulation. Note that the radius of the planet as no direct influence in the calculation of the rocket's movement. It is purely for determining how close a rocket can get to a planet.

As the simulation advances, each agent keeps track of the closest distance it gets to the target. This best distance is used in the fitness function, which is defined as:

$$fitness(R) = \begin{cases} (1 - \frac{d_{\text{best}}}{d_0}) \cdot 100 & \text{if } d_{\text{best}} > r_{\text{target}} \\ 100 & \text{if } d_{\text{best}} \leq r_{\text{target}} \end{cases} \tag{6}$$

The fitness calculates the ratio between the best distance $d_{best}$ of rocket $R$ and the distance between the rocket's starting position and the target, denoted as $d_0$. The inverse formula is then used to represent lower distances as a higher fitness value. In the simulation, reaching the radius of the target threshold ($r_{target}$) stops the rocket's movement, so when this occurs, the fitness is set to the best possible fitness of 100.

The population of rockets are initialized with randomized thrusts, masses, and directions, and use an evolutionary system to explore new genomes in search of a successful combination. After all 5000 frames have been reached, or all rockets have either died or won, the rockets undergo evolution through tournament selection, mutations, and uniform crossovers.

Tournament selection was used to identify the parents that produced the next generation. In each tournament, $k$ random individuals are chosen from the population, and the candidate with the highest fitness score is selected. This is done twice in order to create a pair of parents, which are not removed from the pool when they are used. Next, these two selected genes are put through uniform crossover to produce a singular offspring genome. For each trait in the new genome, a random number between 0 and 1 is chosen. If the number is below the crossover rate, $p_c$, then the offspring takes the value of the current trait from parent 2. Otherwise it inherits from parent 1. Once crossover is finished, it goes through a mutation process. Similar to crossover, a number between 0 and 1 is selected for each trait. If the random number is under the mutation probability, mass and thrust can slightly increase or decrease, and velocity can become a new direction. Mutation is done separately for each trait. The resulting offspring is then added to a new population. This process continues until the offspring population reaches the same size as the previous population. This entire process will continue until the established number of generations is reached.

Every run within this experiment composed of a population size of 100 and tournament size of 4. Based on the best results from pre-experiment testing, $G$ was kept constant at 22 and $dt$ at 0.05. The mutation and crossover rates varied in order to investigate their effects on the behavior within the populations. Every combination of $p_m$ = 0.0, 0.01, 0.05, 0.1 and $p_c$ = 0.0, 0.1, 0.3, 0.5 was tested. Each pair was executed for 20 runs, consisting of 50 generations each. All of this was executed for each of the three maps shown previously. For every generation, the best fitness, average fitness, and success rate (population percentage that reached the target) were recorded, along with the genome for winning individuals every time the last generation (50) was reached.

## IV. Results

### *I. The Dense Environment*

This crowded environment contained many planets with different combinations of mass, ranging from 4000 to 15000 units, and radii, ranging from 5 to 100 units. The visualization can be seen in Fig. 1.

**Env. I: Crossover and Mutation Analysis**

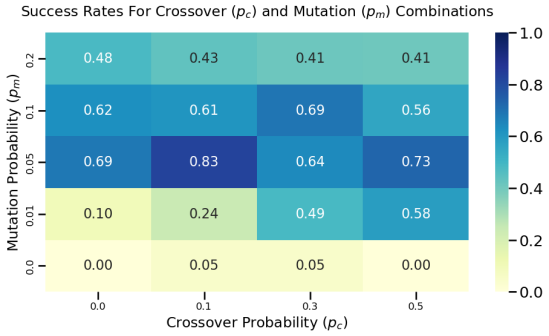



Fig. 2. Mean success rate at generation 50 across all parameter combinations of mutation ($p_m$) and crossover ($p_c$) for Environment I.

The heatmap reveals that the 0.05 mutation rate and 0.1 crossover rate resulted in the highest success rate, with an average of 83% of agents reaching the target when the last generation was reached (50). While that combination is the highest, all simulations with a $p_m$ of 0.05 and 0.1 resulted in better success rates. The worst performance occurred when mutation rate was held at 0.0, which was 0.0% or 5.0% regardless of the crossover rate. Visually, it can also be observed that the success rates fluctuate more drastically vertically (mutation), while the variation horizontally (crossover) is not as significant. This suggests that varying crossover did not have as significant of an effect on performance than altering mutation rate did.

**Env. I: Parameter Effect on Best and Average Fitness**

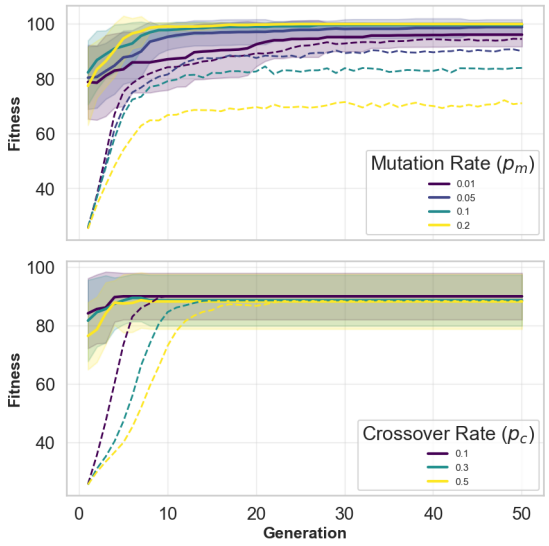



Fig. 3. Evolution of population fitness over 50 generations for Environment I. (a) The top plot displays performance across varying mutation rates with crossover fixed at 0.0. (b) The bottom plot displays varying crossover rates with mutation fixed at 0.0. Solid lines represent the best fitness (shaded by standard deviation), while dotted lines indicate the average fitness of the population.

In Fig. 3(a), the average best finesses within the populations with 0.1 and 0.2 mutation rates were able to reach the perfect value by generation 7. The mean best fitness for a mutation of 0.05 performed slightly worse, but the lowest rate of 0.01 was only able to produce best individuals with an average fitness of 90 by generation 50. However, the overall mean finesses display the opposite behavior, with 0.01 having the highest value. Followed by 0.05 and 0.1, and 0.2 which significantly had a lower average fitness at around 70 when the final generation was reached. All the standard deviations start off larger, and start to shrink as generations increase, but never fully diminishing. When only crossover is utilized in Fig. 3(b), all values showed similar behavior in the average best scoring individual, in which all remained at a value of 90 throughout the entire simulation. Their standard deviations also displayed identical behavior, and stayed at the same size through the plot. While the overall average fitness for every crossover rate eventually converged with the best fitness curves, the speed of convergence varied. The 0.1 rate converged quickest at generation 8, followed by 0.3 at generation 14, and 0.5 at generation 23.

**Env. I: Successful Genomes and Their Trajectories**

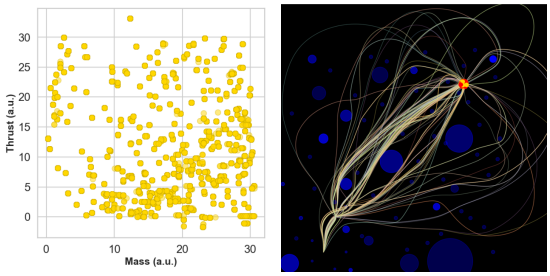



Fig. 4. Genome configurations and successful trajectories for Environment I. (a) Relationship between mass and thrust for all agents reaching the target at generation 50.(b) Representative trajectories of 500 successful genomes selected from the configurations in (a).

In Fig. 4(a), there is a denser accumulation of points in the area with lower thrust and higher mass. This dense region vaguely reveals a linear relationship in the lower right region, starting from roughly (7, 0) to (30, 20). This indicates that the heavier rockets moving slower through the simulation found the most success, and may likely correlate to the thicker region of trajectories in the center of Fig. 4(b). Although, there are also many points scattered throughout the graph, revealing that many other genomes were able to reach the target, but with less efficiency. These outliers may correspond to the more erratic and indirect trajectories that strayed off the central path in Fig. 4(b).

*II. The Symmetric Environment*

The environment features an abundance of planets with uniform masses (2000 units) and radii (60 units), placed in a horizontally symmetric formation, as visualized in Fig. 1.

**Env. II: Crossover and Mutation Analysis**

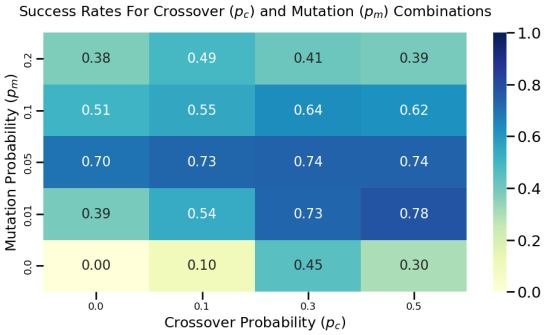



Fig. 5. Mean success rate at generation 50 across all parameter combinations of mutation ($p_m$) and crossover ($p_c$) for Environment II.

The most optimal combination represented in this heat map a crossover rate of 0.5 and a mutation rate of 0.1, which resulted in an average of 78% of rockets reaching the target by the end of 50 generations. Higher success rates are also observed when crossover is high, at 0.3 or 0.5, and mutation rate is mid, at the rates of 0.01, 0.05, and 0.1. The worst performance occurred when both crossover and

mutation were at rates of 0.0. This implies that the most success required some mutation to find a solution, but mostly favored recombining existing traits within the genome.

**Env. II: Parameter Effect on Best and Average Fitness**

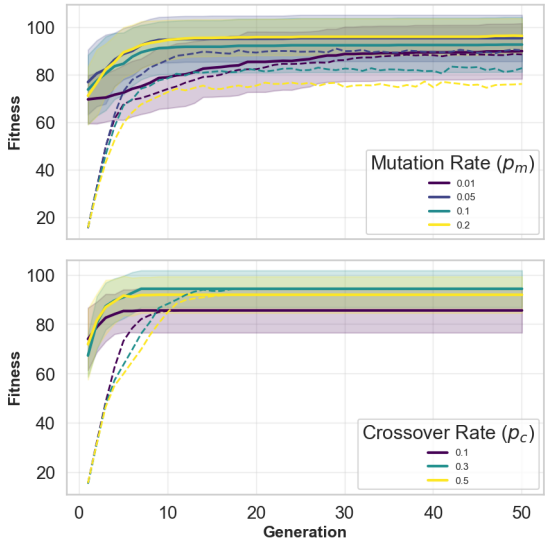



Fig. 6. Evolution of population fitness over 50 generations for Environment II. The top plot displays performance across varying mutation rates with crossover fixed at 0.0, and the bottom plot displays varying crossover rates with mutation fixed at 0.0. Solid lines represent the best fitness (shaded by standard deviation), while dotted lines indicate the average fitness of the population.

Without crossover in Fig. 6(a), the 0.05 and 0.02 mutation rates resulted in the best average best scoring individual, converging to a fitness score slightly above 90. The 0.1 rate performed a little worse as it stayed just below 90. The worst mutation rate was 0.01, where it settled just below the 0.1 curve by generation 50, but had converged much slower compared to the other curves. The average fitness when the mutation rate was 0.05 had the highest results, while 0.2 had the worst. The other two rates start off in the middle of the 0.05 and 0.2 curves, and behave similarly until around generation 20, where 0.1 levels off, but 0.01 continues to gradually increase and reach the same average fitness score as the best performing rate, 0.01, by the last generation. The standard deviations for all remained consistent throughout the plots. In Fig. 6(b), the best fitness curves for all crossover rates converged quickly by generation 5. The rate of 0.3 produced the highest results of an average best fitness of 95, with the 0.5 curve slightly below, and the 0.1 rate at the lowest. Their standard deviations all remain at the same size throughout the graph. The average fitness for all three crossover rates begin to increase at the same speed and converge to their best fitness curves. The rate of 0.1 reaches its best fitness curve at generation 10, whereas 0.3 and 0.5 converge at around generation 17.

**Env. II: Successful Genomes and Their Trajectories**

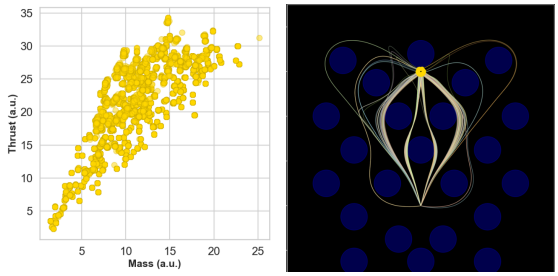



Fig. 7. Genome configurations and successful trajectories for Environment II. (a) Relationship between mass and thrust for all agents reaching the target at generation 50. (b) Representative trajectories of 500 successful genomes selected from the configurations in (a).

The results in Fig. 7(a) show a linear relationship between mass and thrust in this environment, suggesting a mass-to-thrust ratio within the successful genomes. This relationship indicates a balance between navigating through the targets without crashing into the next one. However, as the values increase, there becomes more deviation within the cluster, but no extreme outliers are present. While the ratio is still a main component of success, the system may be more forgiving when higher values are used, allowing for a broader threshold in the greater mass and thrust genomes. In Fig. 7(b), there were four main paths that the rockets took to reach the target. These routes have a symmetric pattern, which mirror the uniform layout of the obstacles. Also, there are a few trajectories that deviate from these central paths.

*III. The Extreme Mass Environment*

Environment III. features a small number of planets placed throughout the map in no particular formation, as seen in Fig. 1. These planets have extremely high masses, ranging from 18000 to 30000 units, and low radii, ranging from 4 to 10 units.

**Env. III: Crossover and Mutation Analysis**

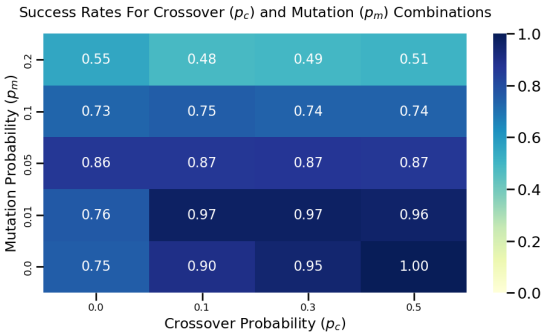



Fig. 8. Mean success rate at generation 50 across all parameter combinations of mutation ($p_m$) and crossover ($p_c$) for Environment III.

The best combination observed in this heatmap is when crossover was 0.5 and mutation was at 0.0. The diagram

showcases a gradient behavior, where success rates start become worse as crossover decreases and mutation increases. Performance was at its worst when mutation was 0.2 with all crossover combinations. Although, at this mutation level, a crossover rate of 0.0 was slightly higher, while still remaining lower than the rest of the heatmap. This implies that the population favored recombining existing traits in the genome, rather than exploring new traits via mutation.

**Env. III: Parameter Effect on Best and Average Fitness**

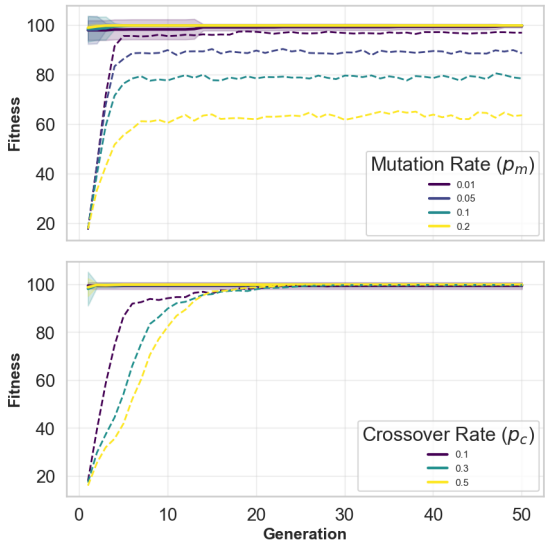



Fig. 9. Evolution of population fitness over 50 generations for Environment III. The top plot displays performance across varying mutation rates with crossover fixed at 0.0, and the bottom plot displays varying crossover rates with mutation fixed at 0.0. Solid lines represent the best fitness (shaded by standard deviation), while dotted lines indicate the average fitness of the population.

The results in Fig. 9(a) reveal that all mutation rates were almost immediately able to produce individuals with a perfect fitness score, as the curves remain at a fitness at or close to 100 for the graph's entirety. The standard deviation for the 0.01 mutation rate is small, but lasts until generation 15, where it shrinks to a very small value, barely detectable. The other curves' deviations diminish completely very early on, around generation 3. However, the average fitness are more differentiable. They all rapidly increased, but level off around generation 5 and remain at the same value until generation 50. The lowest mutation rate, 0.01, had the best average fitness score among the populations, converging at a value very near 100. The 0.05 and 0.1 rates stabilize at average fitness scores of around 90 and 80, respectively. The mutation rate for 0.2 performed the worst, plateauing around 60. The average best fitness score curves for the crossover values in Fig. 9(b) exhibit the same behavior, where all they immediately converge to the highest fitness score. However, there is almost no deviation for these curves in this plot. Additionally, the average fitness scores all converge to a fitness value of 100 around generation 20, but increases at different rates. The crossover rate of 0.1 has the steepest increase until it reaches a fitness value of 90 at generation 5, where it then has a slow increase to 100. 0.3 and 0.5 climb at a slower rate, with 0.3 having a slightly greater slope, but they do not have a dramatic of a peak as 0.1.

**Env. III: Successful Genomes and Their Trajectories**

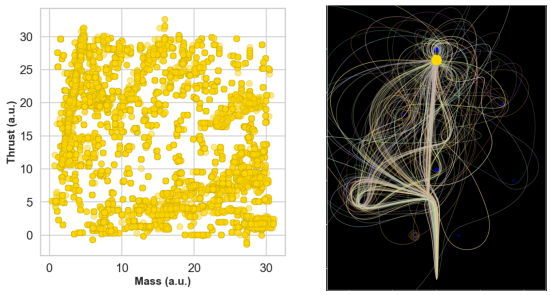



Fig. 10. Genome configurations and successful trajectories for Environment III. (a) Relationship between mass and thrust for all agents reaching the target at generation 50. (b) Representative trajectories of 500 successful genomes selected from the configurations in (a).

In Fig. 10(a), the scatter plot is very crowded with most combinations of high and low mass and thrust resulting in a successful genome combination. However, the center of the plot is slightly more sparse, and the low mass, high thrust and high mass, low thrust regions are slightly more dense. This suggests that better performance occurred when mass and thrust were at extreme values. Fig. 10(b) reveals that there were a wide variety of paths to reach the target. Although, there is a thicker trajectory present, which may correspond to the more saturated parts of the scatter plot.

## V. Discussion, Conclusion, and Future Work

*Mutation Vs. Crossover*

All three environments favored different combinations of mutation and crossover in order to have the highest success rate, demonstrating that the layout of the maps played a role in altering the behavior and optimum parameter values within the evolutionary algorithm. Environment I. favored lower crossover and mid to high mutation, Environment II. preferred high crossover and lower mutation, and Environment III. performed best with the highest value of crossover and no mutation, as seen in the heatmap plots in Fig. 2, Fig. 5, and Fig. 8, respectively.

These findings also align with the successful trajectory routes found within the simulations. In Fig. 4(b), Enviroment I. had a main route going through the center of the map, but also had other routes that deviated, but all still were able to reach the target. This implies that the rockets required a decent amount of exploration to find successful routes at the most efficiency, specifically with a medium-valued mutation rate. Although, this map still favored keeping genome combinations together through a lower crossover rate. To reinforce this notion, Fig. 3(a) demonstrates worse fitness averages when mutation was high, even with no crossover. Also, in Fig. 3(b),

the highest value of crossover was the slowest to converge to the best fitness average and with no mutation, all populations in this graph were unable to produce a mean best fitness of 100. Overall, this environment required mutation to alter genomes rather than using the existing traits in the current gene pool.

However, Environment II.'s map, as seen in Fig. 7(b), reveals four main routes with a minimal amount of trajectories that strayed off the dominant paths. This implies that this environment required some exploration to initially find a successful route, but not as much as Environments I. and III., since Environment II. did not have as many possible solutions as the other maps. Also, in Fig. 6(a) and Fig. 6(b), the average fitness curves were able to quickly converge to the best fitness curves for all crossover values when there was no mutation, while all the average fitness curves in the plot with no crossover were not able to do the same. This corresponds to the results from Fig. 5, as the agents prioritized crossover to recombine existing traits that resulted in one of the few solutions, as opposed to exploring different values of thrust and mass through mutation.

Similar to Environment I., Environment III. also has many solutions that resulted in success. However, Environment III. actually performed better with high rates of crossover and low rates of mutation, which is very different from the preferred parameters for Environment I. In Fig. 9(a) and Fig. 9(b), this environment was able to produce a perfect fitness individual almost immediately. This suggests that this map was very lucky in finding a solution randomly, which may be due to the wide variety of successful trajectories shown in Fig. 10(b). This implies that minimal exploration was needed, and the population depended more on rearranging genomes throughout the generations. Another reason that no mutation was advantageous may be due to the intense gravitational field, where one little mutation could cause the rocket to get stuck in a completely different trajectory.

*Genomes and Trajectories*

The different results observed in the scatter plots and successful trajectories verify that different genomes were optimal for the different environments. Fig. 4(a) shows how the rockets in Environment I. with higher masses and lower thrusts found solutions the most frequently, which suggests that the agents relied more on the gravitational pull from the planets, rather than their own propulsion. Although, other combinations of mass and thrust were also able to reach the target, but they represent a minority of the total successful runs. While these instances don't follow a specific trend and may be due to random exploration, they demonstrate how evolutionary algorithms can benefit from utilizing exploration, rather than simply choosing the best individual in evolution.

However, the trends in Environment II. were much more organized. Fig. 7(a) shows a linear relationship between mass and thrust, that slowly starts to deviate more as the values increase. In Fig. 7(b), the planets are aligned in a horizontal symmetric layout, and the resulting trajectories also display symmetry. In Equation 2, the acceleration due to thrust uses $\frac{F_{thrust}}{m_{rocket}}$, which could indicate that the slope of the linear trend in Fig. 7(a) creates a balance in acceleration between the gravity from the planets and the thrust from the rocket. This proportionality is the golden ratio that allows the rockets to perfectly navigate through the gravitational fields, without crashing into another planet or getting pulled into one as it passes. Overall, this symmetrical environment created a degree of order within the genomes, that lead to more organized trajectories.

Environment III. has the most chaotic trajectories, but also has more points on the scatter plot compared to the other environments, meaning that more rockets were successful overall. The scatter plot shows that many different combinations of thrust and mass lead to success, but there is slightly more density with combinations where at least one of mass or thrust was very high or very low. A lower number rockets reached the target when both mass and thrust were at mid-ranged values. It appears as if the best genomes matched the extreme nature of the environment. In Fig. 10(a), the moderately denser path could have resulted from the genomic traits of an individual not being significant enough to pull itself off that main course. From this data, it appears as if the best genomes matched the extreme nature of the environment. Furthermore, I expected this environment to require a precise result that would be harder to produce, due to the large magnitude of gravity acting on the rockets at once. Instead, the results showed a form of controlled chaos and had the most success out of the three environments. I also expected most rockets to shoot off out of frame (which they do at first), but as they evolve to more efficient genomes, the intense gravity acts as a containment where they had enough time to travel around the grid until the target was reached, especially since the planets' radii were so small that crashing into one was unlikely. This may also correlate to higher success rates occurring with no mutation, since the rockets could simply explore in the grid rather than alter their genomes to achieve a different trajectory. It may also relate to their ability to reach perfect fitness (find the target) at early generations.

*Final Conclusions*

Overall, the behavior of the evolutionary algorithm was shaped by the environment's layout, including the influence of mutation and crossover, and the optimal genomes that produced successful trajectories. The populations were able to become increasingly more successful throughout generations, and could find multiple paths that could reach the target that are unable to be pinpointed simply by just looking at the gravitational field. The trajectory graphs also showcase the existence of local minima, but this experiment aimed, and succeeded, to find different viable paths, rather than converge to a singular trajectory. This experiment demonstrated that even with a simple setup, complex and interesting behaviors could emerge, all based on the environment that the algorithm was tested on. In a much broader sense, these results illustrate how

intricate patterns can emerge from even the most simplistic underlying logic.

*Future Work*

In this experiment, I only defined fitness as the distance from a rocket to the target, which evolved the rockets to simply find any viable path to the target. For future work, I would make the fitness evaluate factors such as time or fuel usage, which will converge the rockets to less paths, in order to better optimize the solutions. Additionally, the current genome consists of a static mass, an initial velocity/direction, and thrust, which always works in the same direction that the rocket is moving. Therefore, I could expand the genome to include more traits, such as the size or shape of a rocket, fuel consumption/variable mass (burning fuel could decrease mass), and thrust angle. To add, this simulation randomized the genome traits based on starting constraints, where the ranges for mass was 1 to 30, for thrust was 0 to 30, and for initial velocity was -1.5 to 1.5 for both the x and y directions. I would be interested to see if expanding these ranges would produce different outcomes, or if the populations would still display the same patterns and behaviors analyzed in this lab. However, I am wondering if even with these simple rules, the populations could eventually converge to one solution, given enough generations. For this simulation, 50 generations proved to not be enough time for populations to evolve toward one trajectory. Therefore, for future work, I would be interested to see if the simulation used in this report could eventually produce a singular, optimal trajectory path.

Besides the genetic algorithm itself, some future work could integrate a more complex environment. For instance, it would be interesting to add dynamic environments, such as moving planets. Finally, this simulation could be scaled to real-world applications, such as Earth and Moon trajectories, by incorporating better proportionality within the masses, sizes, and orbital distances of the planets.

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